

Indian Statistical Institute

Data Analysis with SPSS

by

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SL No.	Topics
1	Fundamentals of Statistics
2	Test of Hypothesis
3	Analysis of Variance
4	Chi Square Test
5	Regression Analysis

FUNDAMENTALS
of
STATISTICS

FUNDAMENTALS OF STATISTICS

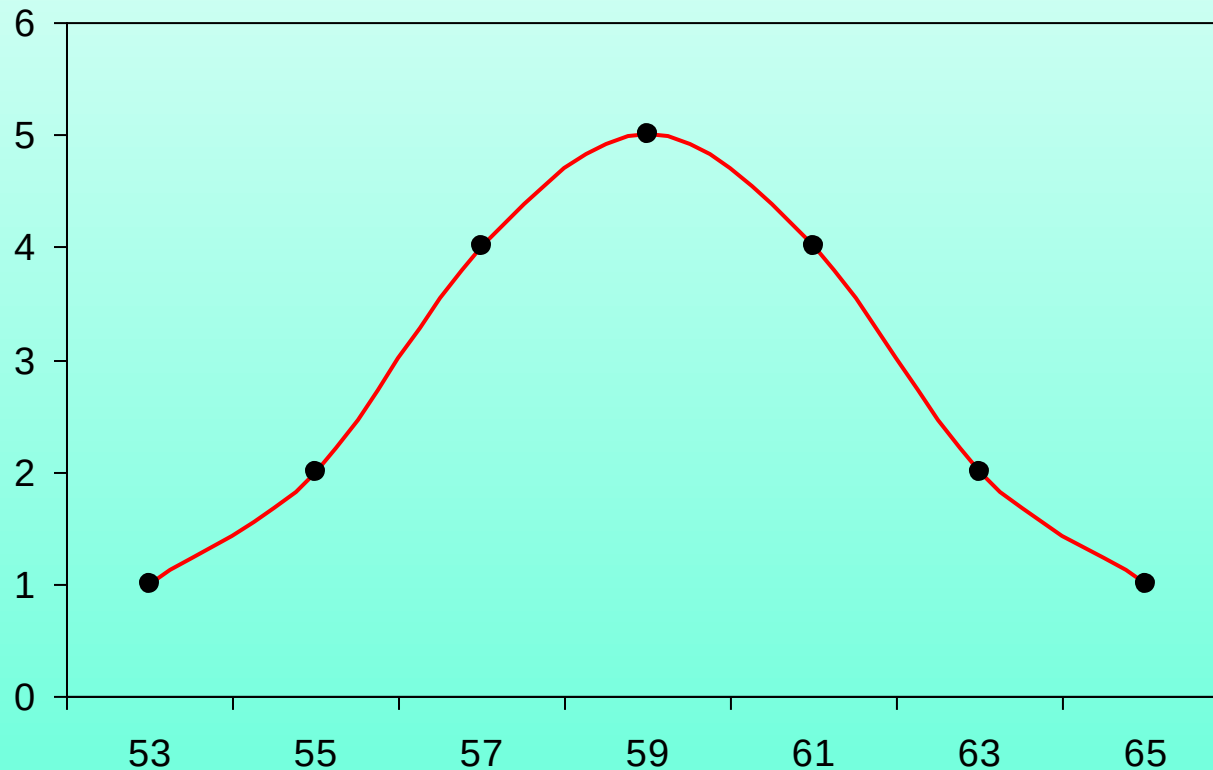
Types of Data

- Continuous or Numeric data (e.g. - Hardness, Strength, Weight, Diameter)
 - Physically measuring the characteristic
 - Generally using an instrument
 - Assigning an unique value to each item
 - Measurable

- Discrete or Nominal data (e.g. - Sex, Shade variation, Surface defects)
 - Classifying the items into different groups based on some criteria
 - No physical measurement is involved
 - All the items classified into a group will have same value
 - Categorical

FUNDAMENTALS OF STATISTICS

Data Summarization



- Summary:**
1. Centre of the Data (Mean, Median, Mode)
 2. Spread or Variation in the Data (Range, Standard Deviation)
 3. Shape of the Curve (Skewness & Kurtosis)

FUNDAMENTALS OF STATISTICS

Continuous Data: Measure of Centre Value

Mean:

Numerical value indicating the central value of data

Sum of all data / Total number of data

Median:

Middle most Value

Value which divides data arranged in ascending or descending order into two equal halves

Mode:

The value which occurs maximum number of times in the data

FUNDAMENTALS OF STATISTICS

Continuous Data Summarization: Measure of Variation or Spread

Range: Definition $R = \text{Maximum value} - \text{Minimum Value}$

Issues:

It depends only on extreme values. Hence affected by outliers

Standard Deviation: Definition

Square root of the average squared deviation from mean

Indicates On an average how much each value is away from the Mean

Graphical representation of frequency table: Bar Charts, Pie Charts, Histogram

FUNDAMENTALS OF STATISTICS

Data Summarization

Example:1 The monthly credit card expenses of an individual in 1000 rupees is given below. Kindly summarize the data

Month	Credit Card Expenses	Month	Credit Card Expenses
1	55	11	63
2	65	12	55
3	59	13	61
4	59	14	61
5	57	15	57
6	61	16	59
7	53	17	61
8	63	18	57
9	59	19	59
10	57	20	63

FUNDAMENTALS OF STATISTICS

How to do it in SPSS? Open SPSS

File -> Open Data -> File Name -> Files of Type (Excel) -> OK

Analyze -> Descriptive Statistics -> Frequencies -> Variable

-> Statistics -> Charts -> OK

•Statistics

•Credit Card Expenses

•Mean 59.20

•Median 59.00

•Mode 59

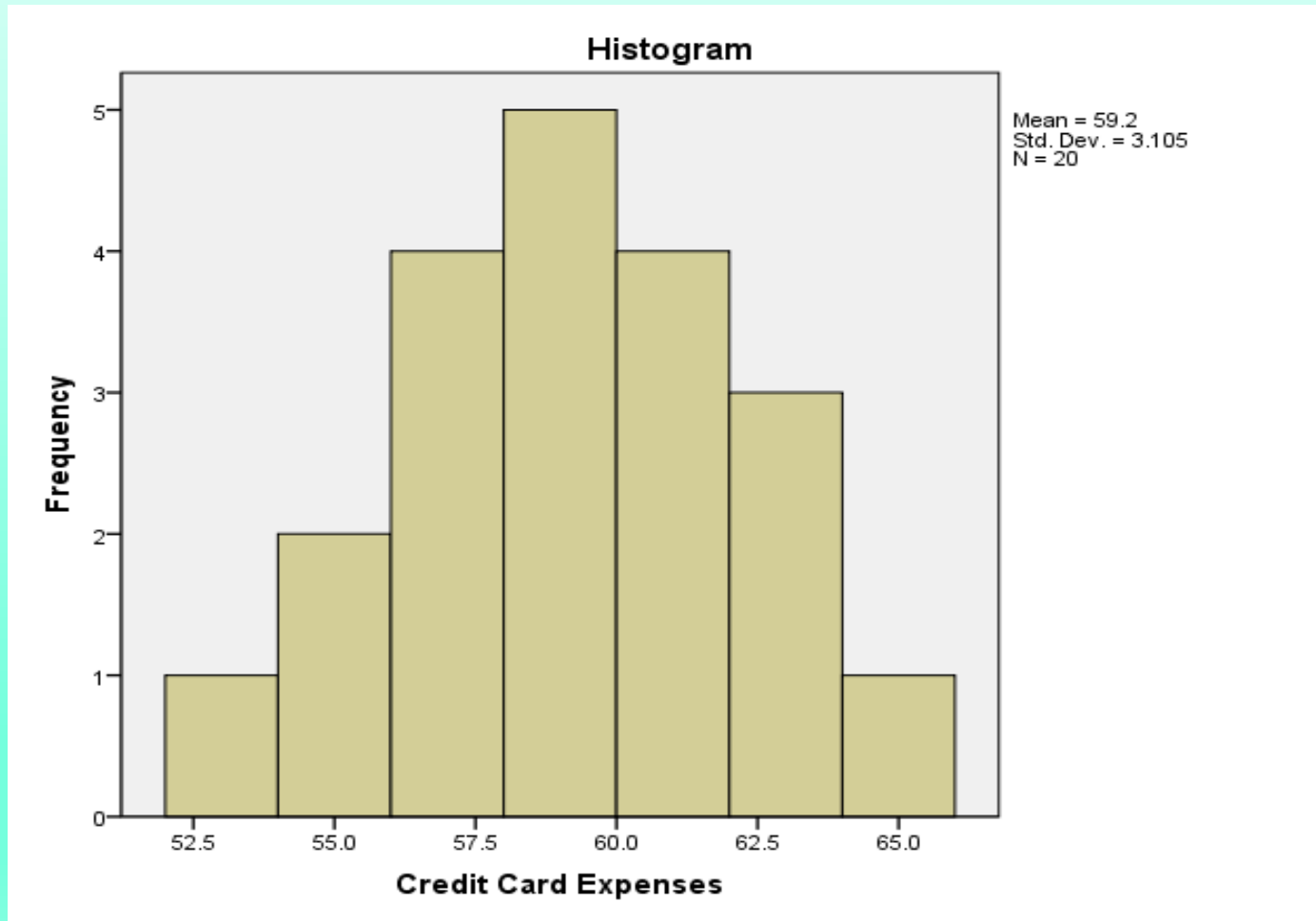
•Std. Deviation 3.105

•Skewness -.090

•Kurtosis -.437

•Range 12

•Histogram shows (Next Page), the data is Normally Distributed.

FUNDAMENTALS OF STATISTICS

**TEST
of
HYPOTHESIS**

TEST OF HYPOTHESIS

Types of Tests:

- Mean equal to a specified value
- Two Means are equal or not
- Two Variances are equal or not
- Proportion equal to a specified value
- Two Proportions are equal or not

TEST OF HYPOTHESIS

Assumptions of Statistical Hypothesis Testing

- the level of measurement of the variable
- the method of sampling
- the shape of the population distribution
- the sample size
- There are actually a variety of different types of hypothesis tests so the specific assumptions will vary for specific tests. However, all hypothesis tests assume probability sampling.

If we wish to test hypotheses about means, as in the example in the book, we have to be sure:

- The sample was randomly selected. This is the case for price data.
- The variable is interval-ratio level. This is true for price.
- Either that the variable is normally distributed in the population or that the sample size is larger than 50. We do not know if prices are normally distributed, but the sample size is greater than 50, so "we know, based on the central limit theorem, that the sampling distribution of the mean will be approximately normal"

TEST OF HYPOTHESIS

Null Hypothesis:

Hypothesis to be checked

Denoted by H_0 :

Example

To check Mean is equal to say 50

Null Hypothesis : $H_0: \text{Mean} = 50$

TEST OF HYPOTHESIS

Alternative Hypothesis:

Hypothesis to be accepted when null hypothesis is not true

Denoted by H_1 :

Example

To check Mean is equal to say 50

There are 3 options:

Alternative Hypothesis : $H_1: \text{Mean} \neq 50$ (2-tail)

$H_1: \text{Mean} > 50$ (1-tail)

$H_1: \text{Mean} < 50$ (1-tail)

TEST OF HYPOTHESIS

Methodology: To Test Mean = Specified Value

Suppose we want to test whether mean of the following data is 5

4	4	5	5	6
5	4.5	6.5	6	5.5

Calculate the Mean of the Sample, $\bar{x} = 5.15$

Check whether $\bar{x} = \text{specified value } 5$

or $\bar{x} - \text{specified value} = \bar{x} - 5 = 0$

If $\bar{x} - 5$ is close to 0

then conclude $\bar{x} = 5$

else $\bar{x} \neq 5$

In this example

$\bar{x} - 5 = 0.15$, Is 0.15 is small enough to conclude $\bar{x} = 5$

TEST OF HYPOTHESIS

Methodology: To Test Mean = Specified Value

Consider another set of data. Check whether mean is 500

400	400	500	500	600
500	450	650	600	550

Mean of the Sample, $\bar{x} = 515$

$$\bar{x} - 500 = 515 - 500 = 15$$

Can we conclude $\bar{x} \neq 500$ since $\bar{x} - 500 = 15 > 0.15$?

If $\bar{x} = \text{specified value}$ then $\bar{x} \times 100$ should be equal to $\text{specified value} \times 100$

Conclusion:

Difficult to say $\bar{x} = \text{specified value}$ by looking at $\bar{x} - \text{Specified value}$ alone

TEST OF HYPOTHESIS

Methodology: To Test Mean = Specified Value

To avoid the scale problem

Test statistic is calculated by standardizing (xbar - specified value) by dividing it with a function of standard deviation

To test Mean = Specified value

$$\text{Test Statistic } t = (\text{Mean} - \text{Specified value}) / (\text{SD} / \sqrt{n})$$

If Test Statistic is close to 0, conclude that Mean = Specified value

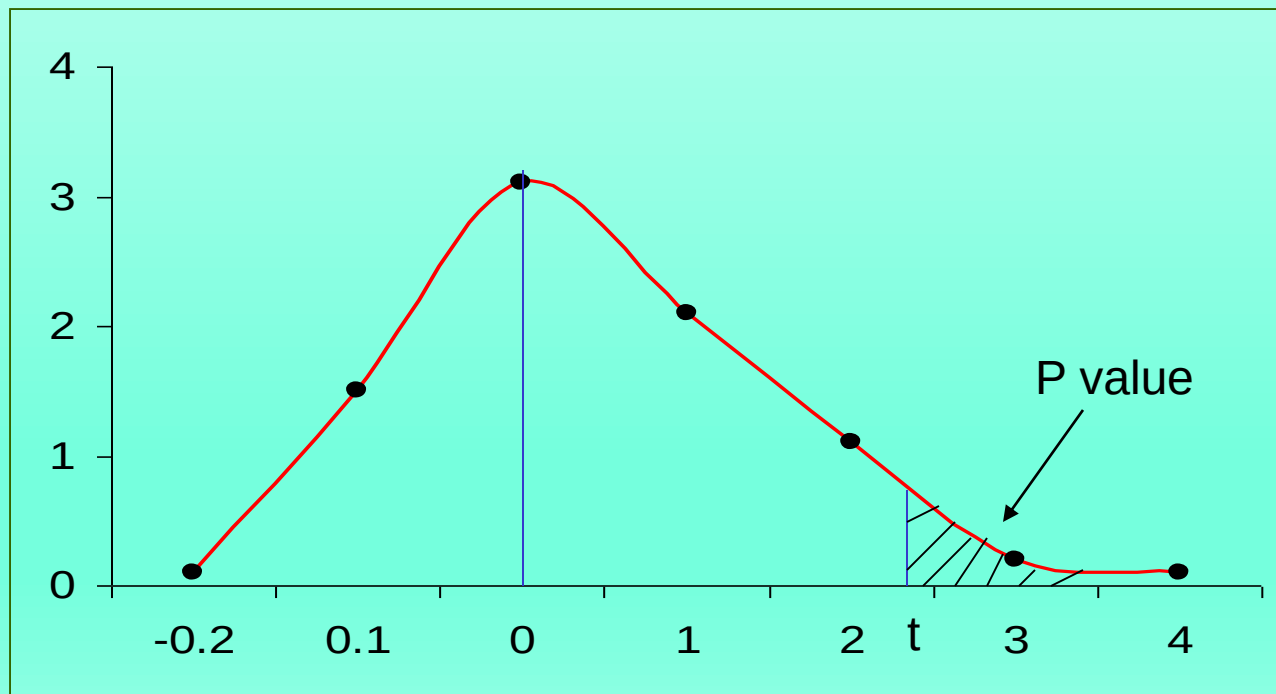
To check Test Statistic is close to 0, find out p value from the distribution of test statistic. P value < 0.05, we reject the null hypothesis.

TEST OF HYPOTHESIS

Methodology: To Test Mean = Specified Value

P value

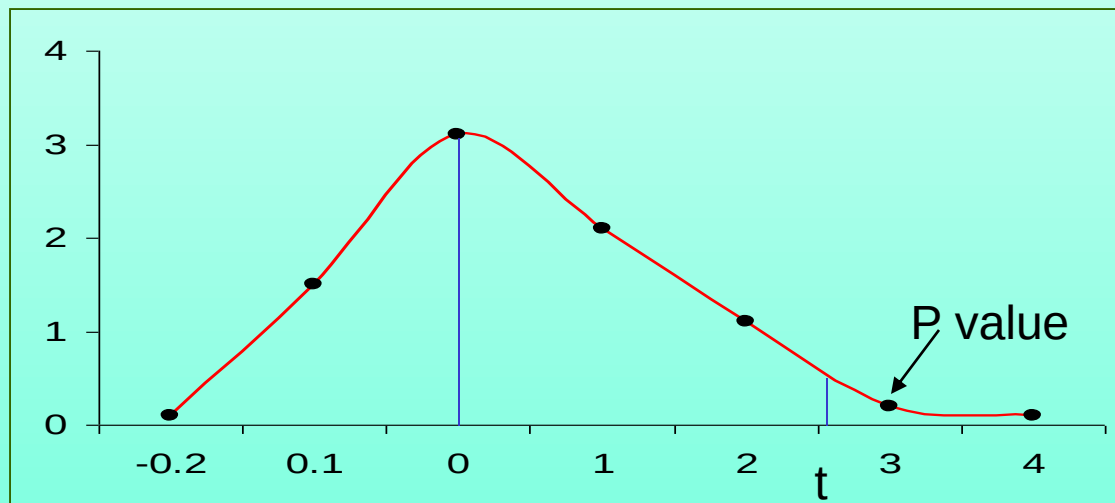
Area above the given test statistic value in the distribution of test Statistics.
If t is the given test statistics, then p is shown in the figure



TEST OF HYPOTHESIS

Methodology: To Test Mean = Specified Value

P value



If Test Statistic t is close to 0 then p will be high

If Test Statistic t is not close to 0 then p will be small

If p is small , $p < 0.05$, conclude that $t \neq 0$

Mean $\neq$ Specified Value

TEST OF HYPOTHESIS

Methodology: To Test Mean = Specified Value

Example: Suppose we want to test whether mean of the following data is 5

4	4	5	5	6
5	4.5	6.5	6	5.5

H0: Mean = 5

H1: Mean $\neq$ 5

Calculate $\bar{x}$ = 5.15

SD = 0.8515

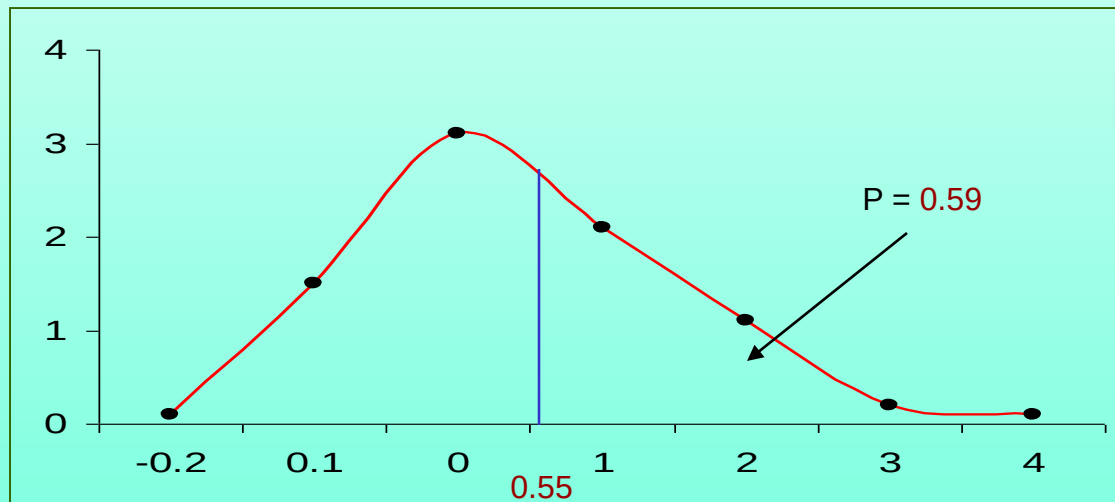
n = 10

Test Statistic $t = (\bar{x} - 5) / (SD / \sqrt{n}) = (5.15 - 5) / (0.8515 / \sqrt{10}) = 0.5571$

TEST OF HYPOTHESIS

Methodology: To Test Mean = Specified Value

$$t = 0.5571$$



$P > 0.05$, hence Mean = Specified value = 5.

Accept H_0 : Mean = 5

TEST OF HYPOTHESIS

To Test Mean = Specified Value:

Example:2 An insurance company claims that on an average it takes only 40 days to process any insurance claim. Based on the data given below, can you validate the claim?

73	19	16	64	28	28	31	90	60	56	31	56	22	
18	45	48	17	17	17	91	92	63	50	51	69	16	17

TEST OF HYPOTHESIS

To Test Mean = Specified Value:

To check Mean is equal to 40

Null Hypothesis : $H_0: \text{Mean} = 40$

Alternative Hypothesis : $H_1: \text{Mean} \neq 40$

Appropriate Test: 2-tailed One Sample t-test

If we are to test $\text{Mean} > 40$, it will be a 1-tail One Sample t-test.

TEST OF HYPOTHESIS

How to do it in SPSS?

Analyse -> Compare Means -> One Sample t-test

Interpretation: If Sig. Value (p-value) < 0.05, we reject H₀.

One-Sample Test						
	Test Value = 40					
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Claim Processing Time	.799	26	.431	3.889	-6.11	13.89

Test: H₀: Mean = 40 Vs. H₁: Mean ≠ 40, p-value is 0.431. So, we accept H₀.

Test: H₀: Mean = 40 Vs. H₁: Mean > 40, p-value is 0.2155. So, we accept H₀.

TEST OF HYPOTHESIS

To Test Two Means are Equal:

Null Hypothesis H_0 : Mean 1 = Mean 2

Alternate Hypothesis H_1 : Mean 1 $\neq$ Mean 2 (2-tail)

H_1 : Mean 1 $>$ Mean 2 (1-tail)

H_1 : Mean 1 $<$ Mean 2 (1-tail)

TEST OF HYPOTHESIS

To Test Two Means are Equal: Methodology

Calculate both Mean 1 & Mean 2

Calculate SD 1 & SD 2

Check Mean 1 = Mean 2

Or Mean 1 - Mean 2 = 0

Calculate Test Statistic t by standardizing (Mean 1 - Mean 2) by a function of SD1 & SD2

$$t = (\text{Mean 1} - \text{Mean 2}) / (S_p \sqrt{((1/n_1) + (1/n_2))})$$

Calculate p from t distribution

If $p > 0.05$ then conclude H_0 : Mean1 = Mean 2 is true

TEST OF HYPOTHESIS

To Test Two Means are Equal:

Example:3 A petroleum company have developed a new formulation for gasoline. Road octane number is an important characteristic of gasoline. 10 observations on road octane number from both formulations is given below: Check whether the means of road octane number for both formulations are same?

Formulation	RON	Formulation	RON
Old	89.5	New	89.5
Old	90	New	91.5
Old	91	New	91
Old	91.5	New	89
Old	92.5	New	91.5
Old	91	New	92
Old	89	New	92
Old	89.5	New	90.5
Old	91	New	90
Old	92	New	91

TEST OF HYPOTHESIS

How to do it in SPSS?

Analyze -> Compare Means -> Independent Samples T-Test

(Note: “Independent Samples T-test means that there are two groups, and we are comparing the means of the two groups.)

- Make the variable for which you want to compute a mean the “Test Variable”
- Make the variable that defines the two groups the “Grouping Variable”
- Click “Define Groups”, then enter the value that each group has on the “Grouping Variable”

Group Statistics

Formulation		N	Mean	Std. Deviation	Std. Error Mean
RON	Old	10	90.700	1.1595	.3667
	New	10	90.800	1.0328	.3266

TEST OF HYPOTHESIS

How to do it in SPSS?

Output:

Interpretation: If Sig. Value (p-value) < 0.05, we reject H0.

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
RON	Equal variances assumed	.239	.631	-.204	18	.841	-.1000	.4910	-1.1316	.9316
	Equal variances not assumed			-.204	17.764	.841	-.1000	.4910	-1.1326	.9326

To find the p-value for the hypothesis test for the difference in means, look in the column labeled “Sig. (2-tailed)” in the “t-test for Equality of Means” section.

Note that since the p-value given by SPSS is 2-tailed, you need to divide it in half for a 1-tailed test.

TEST OF HYPOTHESIS

How to check Histogram of Two Groups (Gr-1: Old, Gr-2: New)

Example:3 A petroleum company have developed a new formulation for gasoline. Road octane number is an important characteristic of gasoline. 10 observations on road octane number from both formulations is given below: Check whether the means of road octane number for both formulations are same?

Formulation	RON	Formulation	RON
Old	89.5	New	89.5
Old	90	New	91.5
Old	91	New	91
Old	91.5	New	89
Old	92.5	New	91.5
Old	91	New	92
Old	89	New	92
Old	89.5	New	90.5
Old	91	New	90
Old	92	New	91

TEST OF HYPOTHESIS

When you have data of different groups together. Split the data & then calculate statistical measures.

How to do it in SPSS:

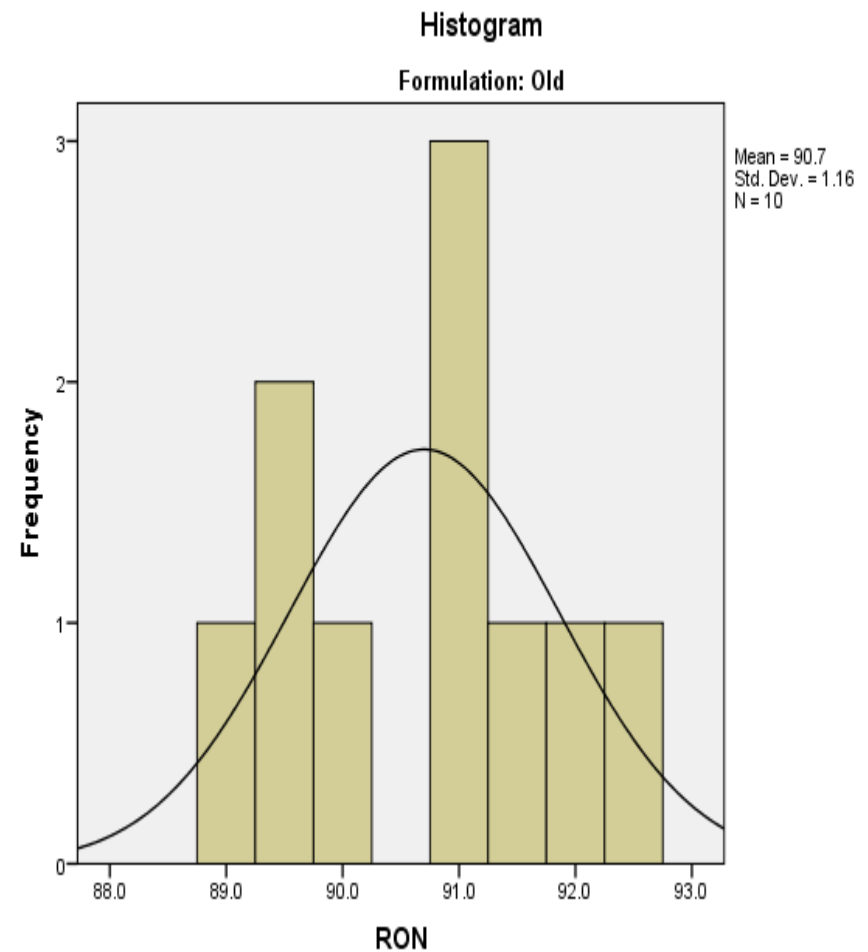
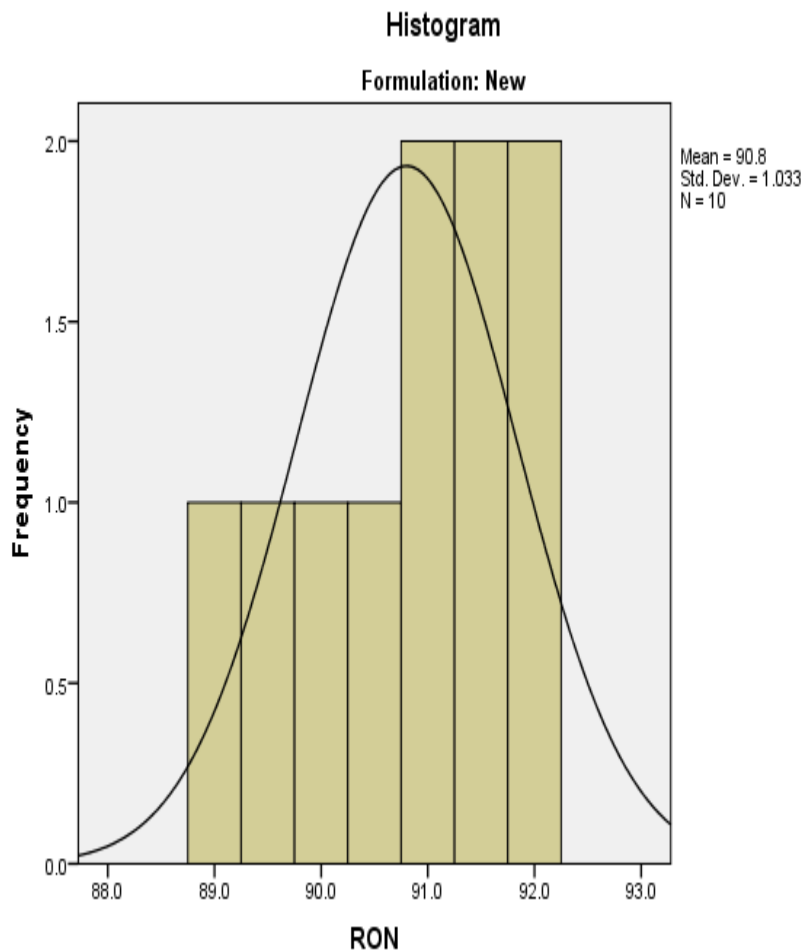
Data-> Split Files -> Organize output by Groups -> Groups Based on Formulation -> Ok

Analyze -> Descriptive Statistics -> Frequencies -> Variable

-> Statistics -> Charts -> Show Normal Curve on Histogram -> OK

TEST OF HYPOTHESIS

SPSS Output: Data do not follow Normal in both the cases.



TEST OF HYPOTHESIS

To Test Two Variances are Equal: Methodology

Null Hypothesis

$$H_0: S_1^2 = S_2^2$$

Alternate Hypothesis

$$H_1: S_1^2 \neq S_2^2$$

Calculate standard deviations of both the samples S_1 & S_2

Calculate Test Statistic $F = S_1^2 / S_2^2$

If F is close to 1, then $S_1^2 = S_2^2$

Calculate p from F distribution.

If $p > 0.05$, then $F = 1$

Accept $H_0: S_1^2 = S_2^2$

TEST OF HYPOTHESIS

To Test Two Variances are Equal:

Example:4 A petroleum company have developed a new formulation for gasoline. Road octane number is an important characteristic of gasoline. 10 observations on road octane number from both formulations is given below: Check whether both formulations have same consistency with respect to RON?

group	RON	group	RON
Old	89.5	New	89.5
Old	90	New	91.5
Old	91	New	91
Old	91.5	New	89
Old	92.5	New	91.5
Old	91	New	92
Old	89	New	92
Old	89.5	New	90.5
Old	91	New	90
Old	92	New	91

TEST OF HYPOTHESIS

Levene's Test of Equal Variances:

First you have to transform the group variable, taking Old = 0, New = 1 (do this step in excel file, then open that file to SPSS) to do One-way ANOVA.

How to do it in SPSS?

Analyze -> Compare Means -> One-way ANOVA-> Dependent List (Ron) -> Factor (group) -> Options -> Homogeneity of Variance Test -> Ok

Test of Homogeneity of Variances			
RON			
Levene Statistic	df1	df2	Sig.
.239	1	18	.631

TEST OF HYPOTHESIS

SPSS Output:

ANOVA

RON

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.050	1	.050	.041	.841
Within Groups	21.700	18	1.206		
Total	21.750	19			

Conclusion: Here variances can be assumed to be equal since p-value is > 0.05.

Note: Levene's test tests the null hypothesis that the **population variances** are equal (called **homogeneity** of variance or **homoscedasticity**).

TEST OF HYPOTHESIS

Exercise: The data of 30 customers on credit card usage in INR1000, sex (1: male, 2: female) and whether they have done shopping or banking (1: yes , 2: no) with credit card are given in table below.

1. Check whether the average credit card usage is same for both sex?
2. Check whether the average credit card usage is same for those who do shopping with credit card and those who don't do shopping?
3. Check whether the average credit card usage is same for those who do banking with credit card and those who don't do banking?

TEST OF HYPOTHESIS

Discrete Data: To Test Proportion is equal to Specified Value

Null Hypothesis $H_0: p = \text{Specified Value}$

Alternate Hypothesis $H_1: p \neq \text{Specified Value}$

$H_1: p > \text{Specified Value}$

$H_1: p < \text{Specified Value}$

TEST OF HYPOTHESIS

To Test Proportion is equal to a Specified Value: Methodology

Calculate sample proportion p

Calculate sample standard deviation $\sqrt{p(1-p)/n}$

Check $p = \text{Specified Value}$

Or $p - \text{Specified Value} = 0$

Calculate Test Statistic z by standardizing $(p - \text{Specified Value})$ by SD

$$z = (p - \text{Specified Value}) / \sqrt{p(1-p)/n}$$

Calculate p from z distribution

If $p > 0.05$ then conclude $H_0: p = \text{Specified Value}$ is true

TEST OF HYPOTHESIS

To Test Proportion is equal to a Specified Value

Example:5

A supply chain company claims that they deliver 98% of shipments without any damage. Based on the shipment data, validate the claim?

Formulation

Null Hypothesis $H_0: p = 0.98$

Alternate Hypothesis $H_1: p \neq 0.98$

First you have to transform the group variable, taking Damaged = 0, Good = 1 (do this step in excel file, then open that file to SPSS)

TEST OF HYPOTHESIS

How to do it in SPSS?

Analyze -> Descriptive Statistics -> Frequencies -> Ok

Transformed Data

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	14	1.4	1.4	1.4
	1	1009	98.6	98.6	100.0
	Total	1023	100.0	100.0	

You can see the percentage value is 98.6%.

The same can be tested by Chi-Square Test

Analyze -> Nonparametric Tests -> Legacy Dialogs -> Ok

TEST OF HYPOTHESIS

SPSS Output

Test Statistics	
	Transformed Data
Chi-Square	967.766 ^a
df	1
Asymp. Sig.	.000

So, their claim can be accepted.

TEST OF HYPOTHESIS

To Test Two Proportion are equal : Methodology

Null Hypothesis $H_0: p_1 = p_2$

Alternate Hypothesis $H_1: p_1 \neq p_2$

$H_1: p_1 > p_2$

$H_1: p_1 < p_2$

TEST OF HYPOTHESIS

To Test Two Proportion are equal : Methodology

Calculate sample proportions p_1 and p_2

Check $p_1 = p_2$

Or $p_1 - p_2 = 0$

Calculate $p = (n_1p_1 + n_2p_2) / (n_1 + n_2)$

Calculate Test Statistic z by standardizing $(p_1 - p_2)$ as

$$z = (p_1 - p_2) / \sqrt{p(1-p)[(1/n_1) + (1/n_2)]}$$

Calculate p from z distribution

If $p > 0.05$ then conclude $H_0: p_1 = p_2$ is true

ANALYSIS
of
VARIANCE

ANALYSIS OF VARIANCE

One Way Anova : Single Factor Anova

A One-Way Analysis of Variance is a way to test the equality of three or more means at one time by using variances.

Example:

To study **Location of shelf** on **Sales Revenue**

ANALYSIS OF VARIANCE

Example: 6

A supermarket chain suspect the location of shelf where toys are kept will influence its sales volume. The data on sales revenue in lakhs from the toys when they are kept at location inside the store are given in Sales Revenue data. The location is denoted as 1:front, 2: middle & 3: rear. Verify the doubt.

ANALYSIS OF VARIANCE

One Way Anova : Example

Factor: Location(A)

Levels : front, middle, rear

Response: Sales revenue

ANALYSIS OF VARIANCE

One Way Anova : Example

Step 1: Calculate the Sum, Average and Number of Responses for each level of the factor (Technology).

Level 1 Sum(A1):

Sum of all responses when Technology is at level 1 (J2EE)

$$= 1.34 + 1.89 + 1.35 + 2.07 + 2.41 + 3.06$$

$$= 12.12$$

nA1: Number of Responses with Technology is at level 1 (J2EE)

$$= 6$$

ANALYSIS OF VARIANCE

One Way Anova : Example

Step 1: Calculate the Sum, Average and Number of Responses for each level of the factor (Technology).

Level 1 Average:

Sum of all responses when Technology is at level 1 / Number of Responses with Technology is at level 1

$$= A1 / nA1 = 12.12 / 6 = 2.02$$

ANALYSIS OF VARIANCE

One Way Anova : Example

Step 1: Calculate the Sum, Average and Number of Responses for each level of the factor (Technology).

	Level 1 (J2EE)	Level 2 (Informix 4GL)	Level 3 (C ++)
Sum	A1: 12.12	A2: 38.09	A3: 7.09
Number	nA1: 6	nA2: 8	nA3: 4
Average	2.02	4.7613	1.7725

ANALYSIS OF VARIANCE

One Way Anova : Example

Step 2: Calculate the Grand Total (T)

T = Sum of all the Responses

$$= 1.34 + 1.89 + - - - + 1.40 + 1.48 = 57.3$$

Step 3: Calculate the Total Number of Responses (N)

$$N = 18$$

Step 4: Calculate the Correction Factor (CF)

CF = (Grand Total)² / Number of Responses

$$= T^2 / N = (57.3)^2 / 18 = 182.405$$

ANALYSIS OF VARIANCE

One Way Anova : Example

Step 5: Calculate the Total Sum of Squares (TSS)

$$\begin{aligned}\text{TSS} &= \text{Sum of Square of all the Responses} - \text{CF} \\ &= 1.34^2 + 1.89^2 + \dots + 1.40^2 + 1.48^2 - 182.405 \\ &= 72.4918\end{aligned}$$

ANALYSIS OF VARIANCE

One Way Anova : Example

Step 6: Calculate the Between Sum of Square of Factor

$$\begin{aligned}SS_A &= A1^2 / nA1 + A2^2 / nA2 + A3^2 / nA3 - CF \\&= 12.12^2 / 6 + 38.09^2 / 8 + 7.09^2 / 4 - 182.405 \\&= 36.0004\end{aligned}$$

Step 7: Calculate the Within Sum of Square

$$\begin{aligned}SS_e &= \text{Total Sum of Square} - \text{Sum of Square of Factors} \\&= TSS - SS_A = 72.4918 - 36.0004 = 36.4914\end{aligned}$$

ANALYSIS OF VARIANCE

One Way Anova : Example

Step 8: Calculate Degrees of Freedom (Df)

$$\begin{aligned}\text{Total Df} &= \text{Total Number of Responses} - 1 \\ &= 18 - 1 = 17\end{aligned}$$

Between Df

$$\begin{aligned}&= \text{Number of levels of the Factor} - 1 \\ &= 3 - 1 = 2\end{aligned}$$

$$\begin{aligned}\text{Within Df} &= \text{Total Df} - \text{Between Df} \\ &= 17 - 2 = 15\end{aligned}$$

ANALYSIS OF VARIANCE

One Way Anova : Example

Anova Table:

Source	Df	SS	MS	F	F Table	P value
Between	2	36.0004	18.0002	7.3991	3.68	0.0058
Within	15	36.4914	2.4328			
Total	17	72.4918	4.2642			

$$MS = SS / Df$$

$$F = MS \text{ Between} / MS \text{ Within}$$

ANALYSIS OF VARIANCE

One Way Anova : Decision Rule

If $p \text{ value} < 0.05$ of a Factor, then

The Factor has significant effect on the Process Output or Response

Meaning:

When the Factor is changed from 1 level to another level, there will be significant change in the response.

ANALYSIS OF VARIANCE

One Way Anova : Example Result

For Factor Location, $p = 0.006 < 0.05$

Conclusion:

Location has significant effect on Sales Revenue

Meaning:

The sales revenue is not same for different locations like front, middle & rear

ANALYSIS OF VARIANCE

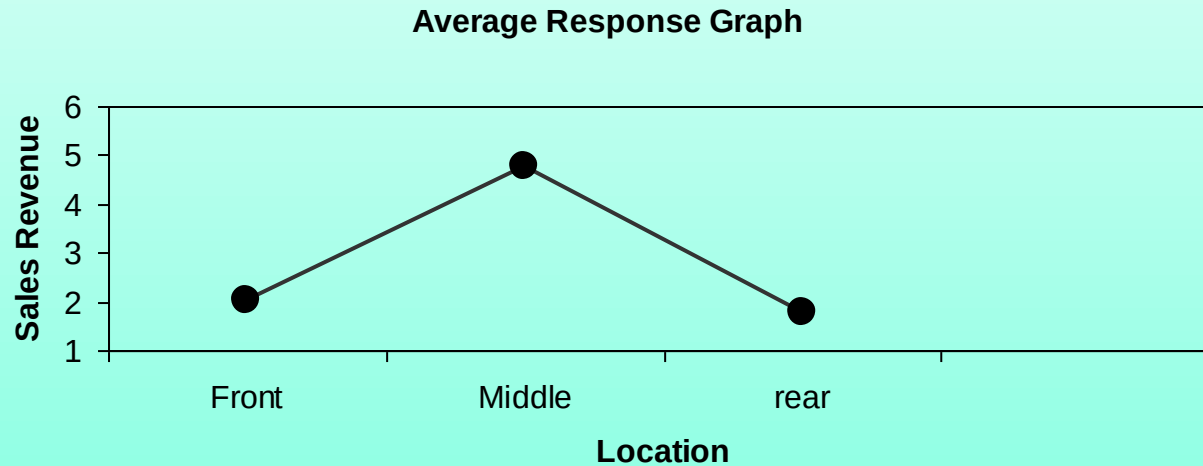
One Way Anova : Example Result

The expected sales revenue for different location under study is equal to level averages.

Location	Expected Sales Revenue
Front	2.0200
Middle	4.7613
rear	1.7725

ANALYSIS OF VARIANCE

One Way Anova : Example Result



ANALYSIS OF VARIANCE

How to do it in SPSS?

Analyse -> Compare Means -> One-way ANOVA -> OK

Options: Mean Plot

Output:

ANOVA					
Sales Revenue					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	36.000	2	18.000	7.399	.006
Within Groups	36.491	15	2.433		
Total	72.492	17			

Conclusion: Location has significant effect on Sales Revenue

CHI SQUARE TEST

CHI SQUARE TEST

Objective:

To test whether two variables are related or not

To check whether a metric is depends of another metric

Usage:

When both the variables (X & Y) are categorical (grouped)

H0: Relation between x & y = 0

H1: Relation between x & y $\neq$ 0

If **p value** < 0.05, then H0 is rejected and x & y are related

CHI SQUARE TEST

Example: 7

A project was undertaken to improve the CSat score of transaction processing. Based on brainstorming, the project team concluded that lack of experience is a cause of low CSat score.

The following data was collected. Analyze the data and verify whether lack of experience is a cause of low CSat score

Experience (Months)	CSat Score				
	VD	D	N	S	VS
0 – 3	50	40	30	10	10
3 – 6	5	30	50	35	7
6 – 9	6	7	30	40	50

Note: Table gives the count of CSat score of 1, 2 etc for each group of agents

CHI SQUARE TEST

Solution:

Step 1: Calculate the Row and column Sum

Experience (Months)	CSat Score					Row Sum
	VD	D	N	S	VS	
0 – 3	50	40	30	10	10	140
3 - 6	5	30	50	35	7	127
6 - 9	6	7	30	40	50	133
Col Sum	61	77	110	85	67	400

CHI SQUARE TEST

Step 2: Calculate Expected Count for each Cell

Expected Count of CSat Score 1 for group 0 – 3 Months experience
 = Expected count of Cell (1,1) = (Row 1 sum x Column 1 sum) / Total
 = (140 x 61) / 400 = 21.4

Table of expected count

Experience (Months)	CSat Score					Row Sum
	VD	D	N	S	VS	
0 – 3	21.4	27	38.5	29.8	23.5	140
3 - 6	19.4	24.4	34.9	27	21.3	127
6 - 9	20.3	25.6	36.6	28.3	22.3	133
Col Sum	61	77	110	85	67	400

CHI SQUARE TEST

Step 3: Take difference between actual count and expected count

For Cell (1,1)

Actual Count = 1

Expected Count = 0.35

Difference = 0.65

Table of Actual Count – Expected count

Experience (Months)	CSat Score				
	VD	D	N	S	VS
0 – 3	28.7	13.1	-8.5	-20	-13
3 - 6	-14.4	5.55	15.1	8.01	-14
6 - 9	-14.3	-19	-6.6	11.7	27.7

CHI SQUARE TEST

Step 4: Calculate $(\text{Actual} - \text{Expected})^2 / \text{Expected}$ for each cell

Table of $(\text{Actual} - \text{Expected})^2 / \text{Expected}$

Experience (Months)	CStat Score				
	VD	D	N	S	VS
0 – 3	38.45	6.32	1.88	13.11	7.71
3 - 6	10.66	1.26	6.51	2.38	9.58
6- 9	10.06	13.52	1.18	4.87	34.50

CHI SQUARE TEST

Step 5: Calculate Chi Square = Sum of all $((\text{Actual} - \text{Expected})^2 / \text{Expected})$

Chi Square calculated = $38.45 + 6.32 + \dots + 34.5$

Chi Square Calculated $\chi^2 = 161.98$

Step 6: Calculate p value

P value = $\text{chidist}(\text{chi Sq}, \text{df})$
= $\text{chidist}(161.98, 8)$
= 0.00

Conclusion:

Since p value $0.00 < 0.05$, there is a significant impact experience has on Csat score

CHI SQUARE TEST

Issues:

- Chi square test only shows whether two variables are related or not
- Degree of association will not be known

Measures of Strength of relationship:

1. Phi (ϕ) Coefficient

$$\phi = \sqrt{\chi^2 / n}$$

Only for 2 x2 tables

2. Cramer V = $\sqrt{(\phi^2 / (\min (\text{rows} - 1) , (\text{cols} - 1)))}$

Phi & V varies from 0 to 1, higher the value better the strength of relation

CHI SQUARE TEST

$$\text{Phi Coefficient} = \sqrt{161.98 / 400} = 0.64$$

Cramer V:

$$\text{Rows} - 1 = 2$$

$$\text{Columns} - 1 = 4$$

$$\text{Cramer V} = \sqrt{(0.64^2 / 2)} = 0.4499 = 44.99\%$$

CHI SQUARE TEST

Tau b

- Varies from -1 to +1
- Close to -1 indicate good negative correlation, Close to +1 indicate good positive correlation & close to 0 indicate no relation
- Used for $n \times n$ tables

Tau c

- Varies from -1 to +1
- Close to -1 indicate good negative correlation, Close to +1 indicate good positive correlation & close to 0 indicate no relation
- Used for $r \times c$ tables (r not equal to c)

CHI SQUARE TEST

How to do it in SPSS?

Analyse -> Descriptive Statistics -> Crosstabs -> Rows & Columns -> Statistics -> OK

**CORRELATION
&
REGRESSION**

CORRELATION & REGRESSION

Correlation:

Correlation analysis is a technique to identify the relationship between two variables.

Type and degree of relationship between two variables.

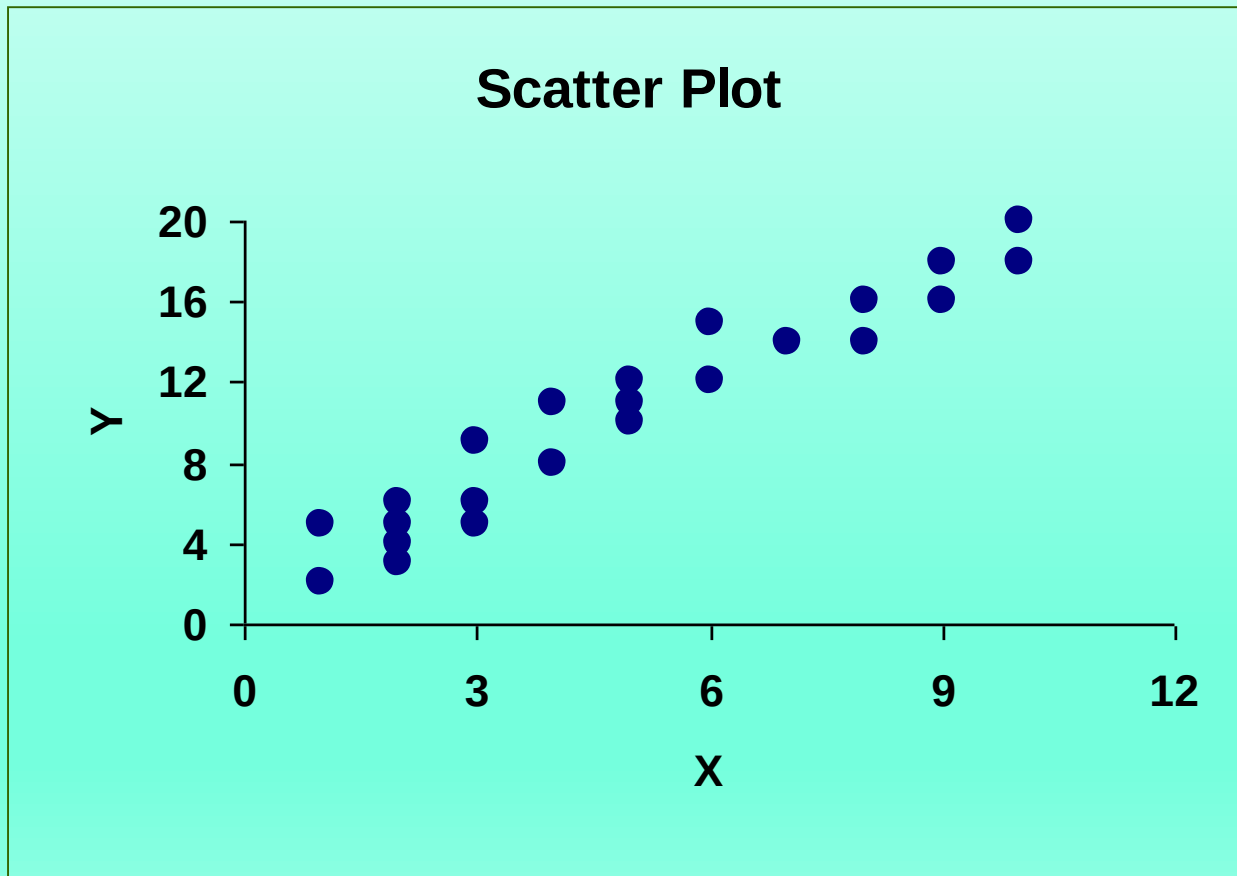
Explore the relationship between the output characteristic and input or process variable.

Output variable : Y : Dependent variable

Input / Process variable : X : Independent variable

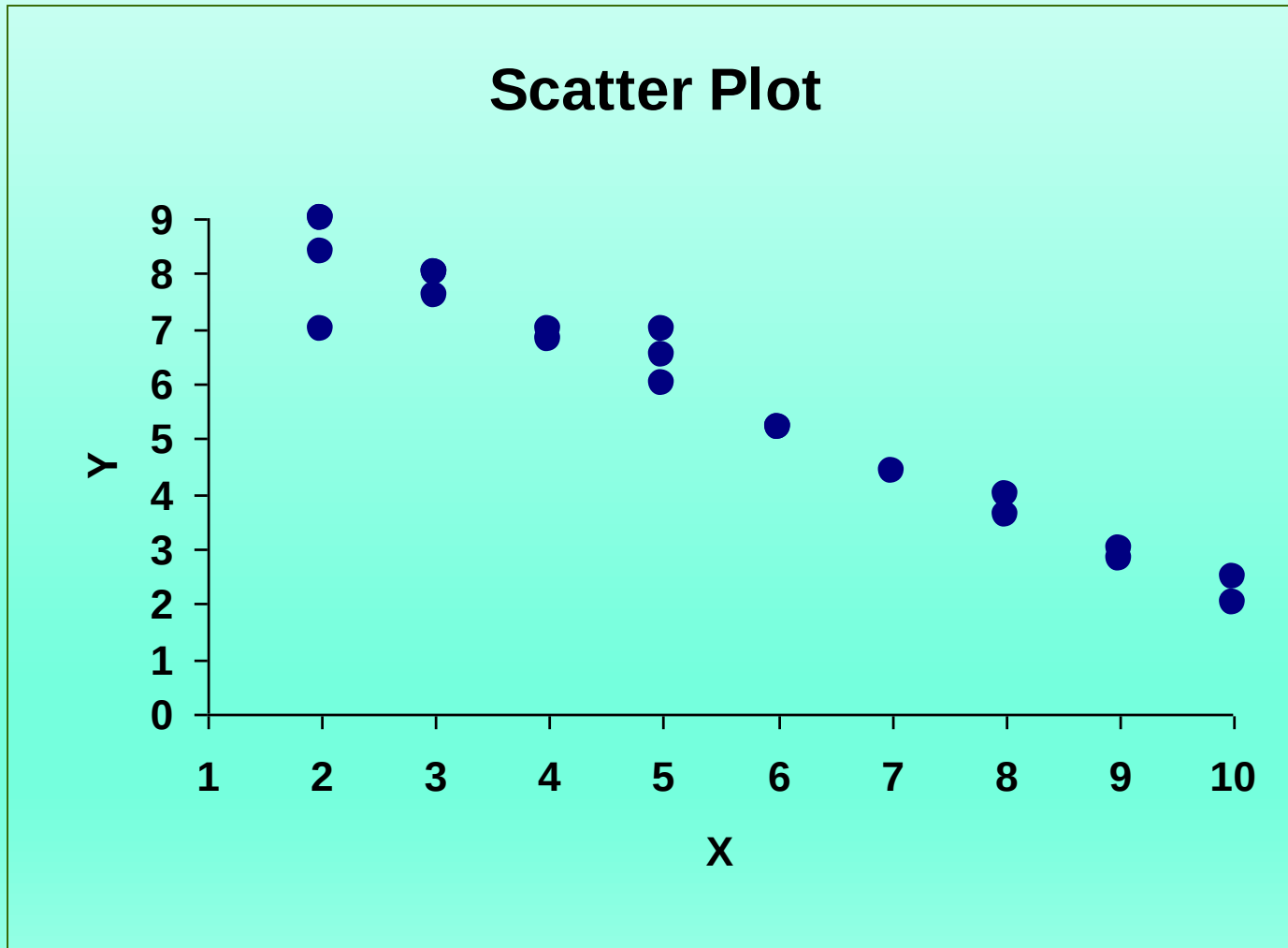
CORRELATION & REGRESSION

Positive Correlation: Y increases as X increases & vice versa



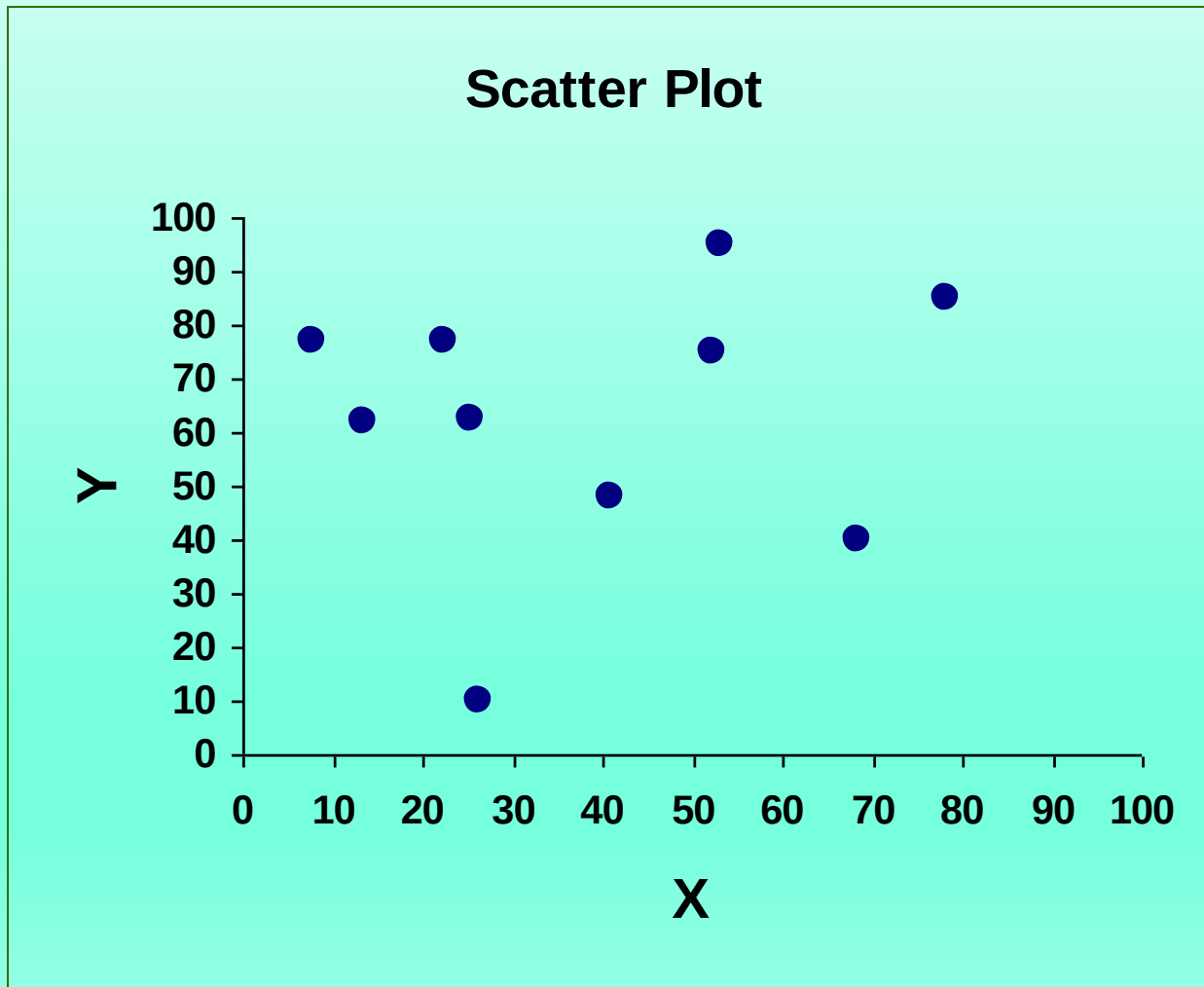
CORRELATION & REGRESSION

Negative Correlation: Y decreases as X increases & vice versa



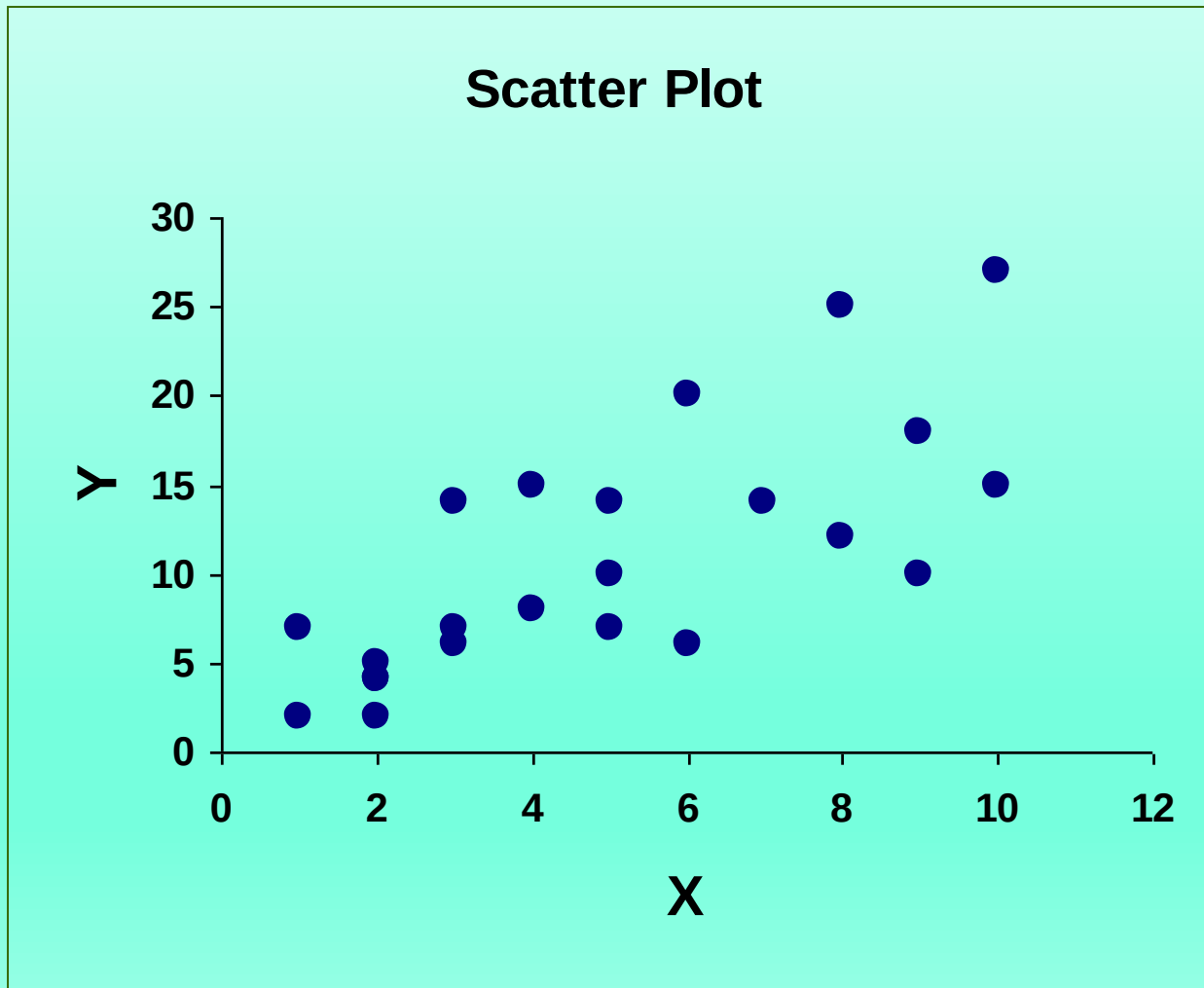
CORRELATION & REGRESSION

No Correlation: Random Distribution of points



CORRELATION & REGRESSION

Is there any correlation ?



CORRELATION & REGRESSION

Measure of Correlation: Coefficient of Correlation

Symbol : r

Range : -1 to 1

Sign : Type of correlation

Value : Degree of correlation

Examples:

$r = 0.6$, 60 % positive correlation

$r = -0.82$, 82% negative correlation

$r = 0$, No correlation

CORRELATION & REGRESSION

Coefficient of Correlation:

In Short

If correlation is positive

S_{xy} will be positive

If correlation is negative

S_{xy} will be negative

CORRELATION & REGRESSION

Coefficient of Correlation:

To avoid scale issues

S_{xy} is divided by $\sqrt{(S_{xx} \cdot S_{yy})}$

$$S_{xy} = \sum (x - \text{Mean } x)(y - \text{Mean } y)$$

$$S_{xx} = \sum (x - \text{Mean } x)^2$$

$$S_{yy} = \sum (y - \text{Mean } y)^2$$

$$\text{Correlation Coefficient } r = S_{xy} / \sqrt{(S_{xx} \cdot S_{yy})}$$

CORRELATION & REGRESSION

Correlation Coefficients: Nominal Data

1. Spearman's rho (ρ)
2. Kendall's Tau (τ)

Varies from -1 to +1

Close to -1 indicate negative correlation

Close to +1 indicate positive correlation

Close to 0 means no correlation

Used for nominal data

CORRELATION & REGRESSION

Regression

Correlation helps

To check whether two variables are related

If related

Identify the type & degree of relationship

CORRELATION & REGRESSION

Regression

Regression helps

- To identify the exact form of the relationship
- To model output in terms of input or process variables

Examples:

Yield = 5 + 3 x Time - 2 x Temperature

$Y = 2 - 5x$

CORRELATION & REGRESSION

Simple Linear Regression: Parameter Estimation

Model: $Y = a + bx$

$$a = \text{Mean } y - b \cdot \text{Mean } x$$

$$b = S_{xy} / S_{xx}$$

Test for Significance (Testing $b = 0$ or not) of relation between x & y

$$H_0: b = 0$$

$$H_1: b \neq 0$$

Test Statistic $t = (b - 0) / (SD / \sqrt{n})$

If $p \text{ value} < 0.05$, then H_0 is rejected & y can be model with x

CORRELATION & REGRESSION

Coefficient of Regression: Measure of degree of Relationship

Symbol : R^2

$$R^2 = SS_R / S_{yy} = b \cdot S_{xy} / S_{yy}$$

$$SS_R = \sum (y_{\text{predicted}} - \text{Mean } y)^2$$

$$S_{yy} = \sum (y_{\text{actual}} - \text{Mean } y)^2$$

Range of R^2 : 0 to 1

If $R^2 > 0.6$, the Model is reasonably good

CORRELATION & REGRESSION

Coefficient of Regression: Testing the significance of Regression

Regression ANOVA

Model	SS	df	MS	F	p value
Regression	SS_R				
Residual	$Syy - SS_R$				
Total	Syy				

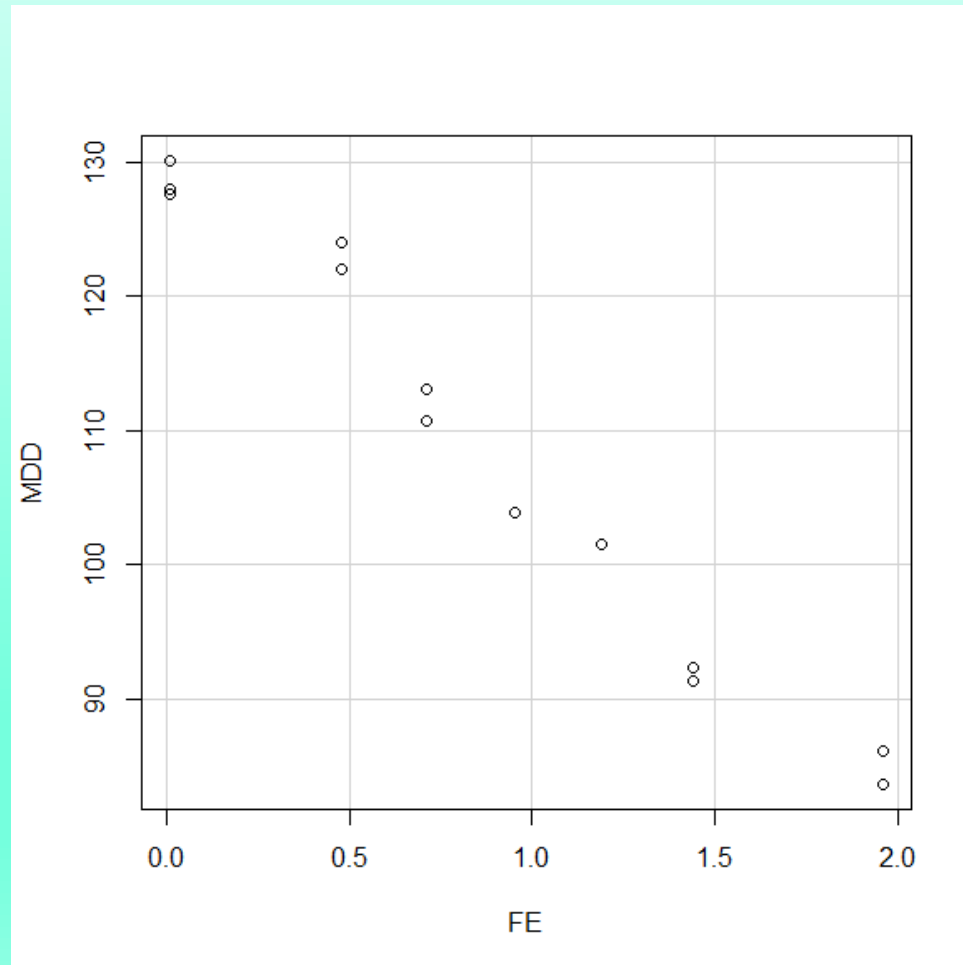
If $p \text{ value} < 0.05$, then the regression model is significant

CORRELATION & REGRESSION

Example:8 Thirteen specimens of 90/10 Cu-Ni alloys, each with a specific iron content, was tested in a corrosion- wheel set up. The wheel was rotated in salt sea water at 30 ft/s for 60 days. The corrosion was measured in weight loss in milligrams/square diameter/day, MDD. The data obtained is given in MDD_Simple_Reg file. Develop a prediction model for MDD in terms of iron content.

CORRELATION & REGRESSION

Step 1: Scatter Plot



CORRELATION & REGRESSION

Step 2: Correlation Matrix

Attribute	FE	MDD
FE	1.000	-0.985
MDD	-0.985	1.000

Remark:

Correlation between Y & X need to be 0.8 to 1 to -0.8 to -1.0

CORRELATION & REGRESSION

Step 3: Regression Output

Statistic	Value	Criteria
R Square	0.9697	> 0.6

Regression ANOVA

Model	SS	df	MS	F	p value
Regression	3293.76669	1	3294	352.3	0.0000
Residual	102.850233	11	9.35		
Total	3396.616923	12			

Criteria: P value < 0.05

CORRELATION & REGRESSION

Step 3: Regression Output

Attribute	Coefficient	Std. Error	Std. Coefficients	Tolerance	t Statistic	p value
FE	-24.01989	1.22368	-18.9759	1	-19.629	0.00
(Intercept)	129.7866	1.33321			97.3491	0.00

Interpretation

The p value for independent variable should be < 0.05

Model: $MDD = -24.020 * FE + 129.787$

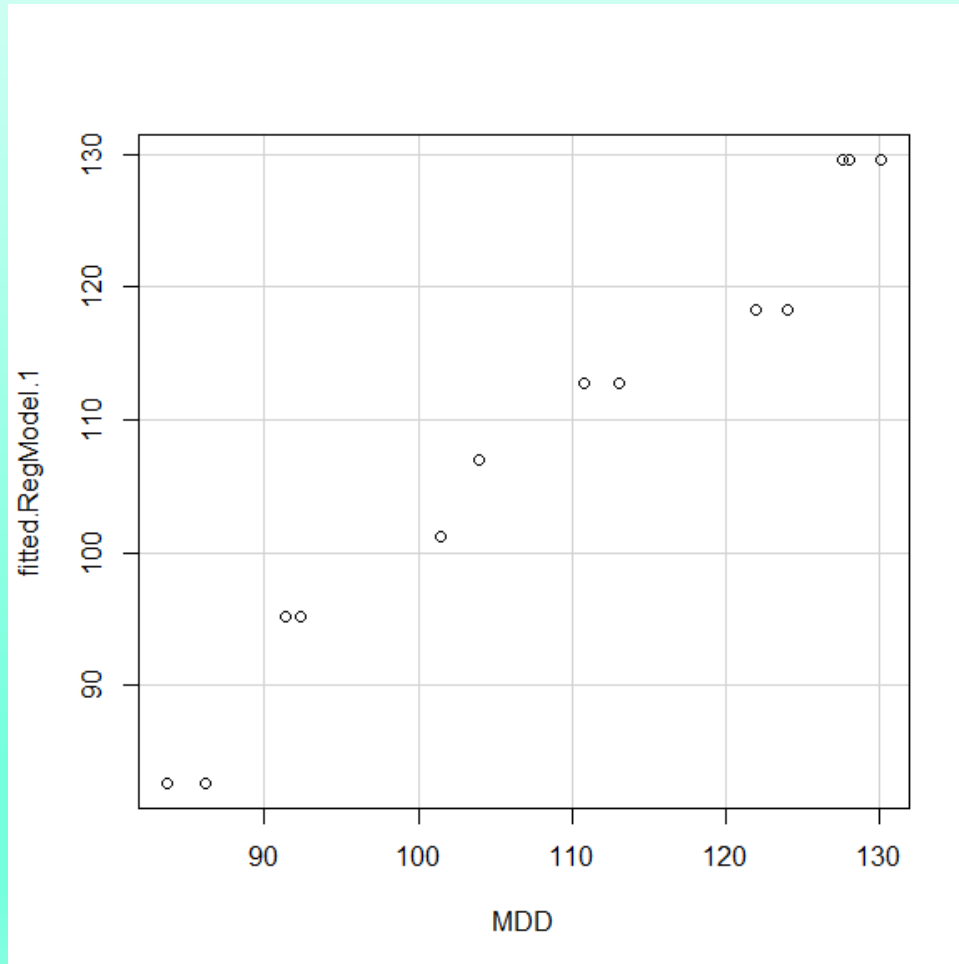
CORRELATION & REGRESSION

Step 4: Model Diagnostics

SL No.	MDD	Fit	FE
1	127.60	129.55	0.01
2	124.00	118.26	0.48
3	110.80	112.73	0.71
4	103.90	106.97	0.95
5	101.50	101.20	1.19
6	130.10	129.55	0.01
7	122.00	118.26	0.48
8	92.30	95.20	1.44
9	113.10	112.73	0.71
10	83.70	82.71	1.96
11	128.00	129.55	0.01
12	91.40	95.20	1.44
13	86.20	82.71	1.96

CORRELATION & REGRESSION

Step4: Model Diagnostics - Model Performance



Scatter Plot: Actual Vs Predicted

CORRELATION & REGRESSION

Step 5: Residual Analysis

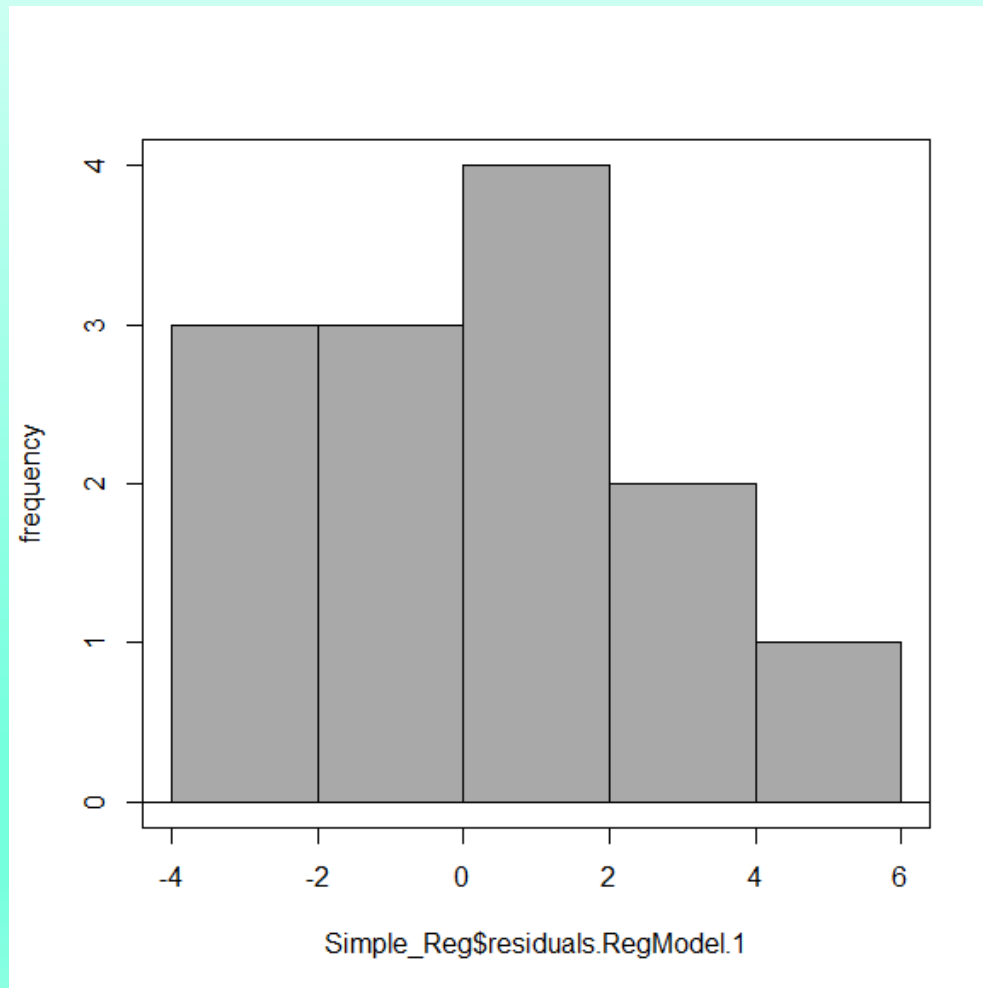
Residual: Actual – Fitted

Standardized Residual: $(\text{Residual} - \text{Mean}) / \text{Standard Deviation of Residuals}$

Observation	Actual MDD	Predicted MDD	Residuals	Standard Residuals
1	127.6	129.55	-1.95	-0.66
2	124	118.26	5.74	1.96
3	110.8	112.73	-1.93	-0.66
4	103.9	106.97	-3.07	-1.05
5	101.5	101.20	0.30	0.10
6	130.1	129.55	0.55	0.19
7	122	118.26	3.74	1.28
8	92.3	95.20	-2.90	-0.99
9	113.1	112.73	0.37	0.13
10	83.7	82.71	0.99	0.34
11	128	129.55	-1.55	-0.53
12	91.4	95.20	-3.80	-1.30
13	86.2	82.71	3.49	1.19
Mean			0.00	0.00
SD			2.93	1.00

CORRELATION & REGRESSION

Step 5: Residual Analysis



Residuals should be normally distributed or bell shaped

CORRELATION & REGRESSION

Step 5: Residual Analysis

Residuals should be normally distributed or bell shaped

Normality Test

H0: Deviation from bell shape = 0

H1: Deviation from bell shape $\neq$ 0

If p value ≥ 0.05 , then H0 is accepted and residuals are normally distributed

Shapiro-Wilk normality Test: MDD data	
W	p value
0.933	0.3733

CORRELATION & REGRESSION

Step 5: Residual Analysis: Detection of Outliers

Observation	Actual MDD	Predicted MDD	Residuals	Standard Residuals
1	127.6	129.55	-1.95	-0.66
2	124	118.26	5.74	1.96
3	110.8	112.73	-1.93	-0.66
4	103.9	106.97	-3.07	-1.05
5	101.5	101.20	0.30	0.10
6	130.1	129.55	0.55	0.19
7	122	118.26	3.74	1.28
8	92.3	95.20	-2.90	-0.99
9	113.1	112.73	0.37	0.13
10	83.7	82.71	0.99	0.34
11	128	129.55	-1.55	-0.53
12	91.4	95.20	-3.80	-1.30
13	86.2	82.71	3.49	1.19
Mean			0.00	0.00
SD			2.93	1.00

Any observation with standardized residuals beyond ± 3 or residuals beyond ± 3 standard deviation

CORRELATION & REGRESSION

How to do it in SPSS?

Graphs -> Chart Builder -> Scatter Plot

Analyse -> Descriptive Statistics -> Explore -> Plots -> Normality plots with tests

Analyse -> Regression -> Linear

CORRELATION & REGRESSION

Multiple Linear Regression

To model output variable y in terms of two or more variables.

General Form:

$$Y = a + b_1X_1 + b_2X_2 + \dots + b_kX_k$$

Two variable case:

$$Y = a + b_1X_1 + b_2X_2$$

CORRELATION & REGRESSION

Example: 9 The effect of sealer plate temperature and sealer plate clearance in a soap wrapping machine affects the % of wrapped bars that pass inspection. The data collected is given in the Mult-Reg_Sealed file. Develop a model for % Sealed Property in terms of clearance and temperature?

Step 1: Correlation Analysis

Attribute	Clearance	Temperature	% Sealed Property
Clearance	1.00	-0.01	0.90
Temperature	-0.01	1.00	-0.05
% Sealed Property	0.90	-0.05	1.00

Correlation between Xs & Y should be high

Correlation between Xs should be low

CORRELATION & REGRESSION

Step 2: Regression Output

Statistic	Value	Criteria
Adjusted R Square	0.7766	> 0.6

Regression ANOVA

Model	SS	df	MS	F	p value
Regression	6797.063	2	3398.531	27.07	0.0000
Residual	1632.08138	13	125.5447		
Total	8429.14438	15			

Criteria: P value < 0.05

CORRELATION & REGRESSION

Step 2: Regression Output – Identify the model

Attribute	Coefficient	Std. Error	Std. Coefficients	Tolerance	t Statistic	p value
Clearance	0.9061	0.1004	0.1296	0.9999	9.0266	0.0000
Temperature	-0.0642	0.0844	-0.0053	0.9999	-0.7604	0.4669
(Intercept)	-67.8844	10.1020			-6.7199	0.0000

Interpretation: Only Temperature is related to % Sealed property as p value < 0.05

CORRELATION & REGRESSION

Step 2: Regression Output – Identify the model

Attribute	Coefficient	Std. Error	Std. Coefficients	Tolerance	t Statistic	p value
Clearance	0.9065	0.05755	0.129664		15.74989	0.00
(Intercept)	-81.6205	9.24078			-8.83264	0.00

Model % Sealed property = 0.906 * Clearance - 81.621

CORRELATION & REGRESSION

Step 3: Residual Analysis

SL No.	Temperature	% Sealed Property	Prediction	Clearance
1	190	35.0	36.22	130
2	176	81.7	76.10	174
3	205	42.5	39.84	134
4	210	98.3	91.51	191
5	230	52.7	67.94	165
6	192	82.0	94.23	194
7	220	34.5	48.00	143
8	235	95.4	86.98	186
9	240	56.7	44.38	139
10	230	84.4	88.79	188
11	200	94.3	77.01	175
12	218	44.3	59.79	156
13	220	83.3	90.61	190
14	210	91.4	79.73	178
15	208	43.5	38.03	132
16	225	51.7	52.53	148

CORRELATION & REGRESSION

Step 3: Residual Analysis: Outlier detection

SL No.	Temperature	% Sealed Property	Prediction	Clearance	Residuals	Std Residuals
1	190	35	36.22	130	-1.22	-0.116
2	176	81.7	76.1	174	5.60	0.534
3	205	42.5	39.84	134	2.66	0.253
4	210	98.3	91.51	191	6.79	0.647
5	230	52.7	67.94	165	-15.24	-1.453
6	192	82	94.23	194	-12.23	-1.166
7	220	34.5	48	143	-13.50	-1.287
8	235	95.4	86.98	186	8.42	0.802
9	240	56.7	44.38	139	12.32	1.174
10	230	84.4	88.79	188	-4.39	-0.418
11	200	94.3	77.01	175	17.29	1.648
12	218	44.3	59.79	156	-15.49	-1.476
13	220	83.3	90.61	190	-7.31	-0.697
14	210	91.4	79.73	178	11.67	1.112
15	208	43.5	38.03	132	5.47	0.521
16	225	51.7	52.53	148	-0.83	-0.079
				Mean	0.000625	
				SD	10.4918333	

CORRELATION & REGRESSION

Step 3: Residual Analysis:

Shapiro-Wilk normality Test: MDD data	
W	p value
0.9471	0.4449

THANK YOU